

16	A
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17	D
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19	C
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	27	D
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14	C
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15	C
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17	D
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18	A
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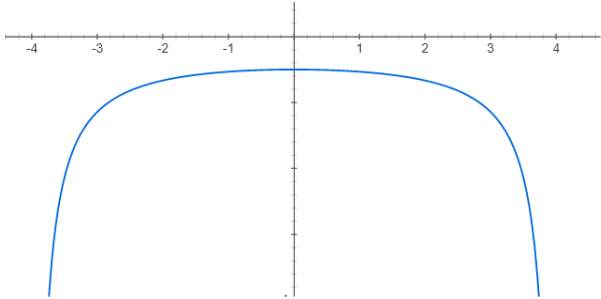
19	A
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13	A
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14	B
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14	B	electric field	magnitude of charge on Y > magnitude of charge on X	AO1
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Question	Marking guidance	Mark	Comments
04.1	Use of $E_{\text{res}} = 2E \cos\theta$ AND use of $E = \frac{Q}{4\pi\epsilon_0 r^2}$ ✓ distance to one charge found (= 5.0 mm) OR $\cos\theta$ OR θ calculated correctly ✓ $Q = -3.2 \times 10^{-12}$ (C) ✓	3	($\cos \theta = 0.6$; $\theta = 53^\circ$) Condone missing minus sign Allow answers between -3.2 and -3.3×10^{-12} (C)

Question	Marking guidance	Mark	Comments
04.2	Any smooth curve that is always negative✓ MP1 AND with a non-zero maximum at y-axis AND asymptotic at 4, -4✓	2	Condone touching x-axis for MP1 Allow 1 for the correct curve reflected in x-axis Expect: 

Question	Marking guidance	Mark	Comments
04.3	Direction is always towards O owtte ✓ Idea that both Q charges will repel m, and the nearer Q will repel more ✓	2	

Question	Marking guidance	Mark	Comments
04.4	Electric potential energy is transferred to kinetic energy (as it approaches the centre) OR EPE decreases and KE increases ✓ Decrease in EPE is equal to the increase in KE ✓	2	Max 1 for answers including idea that EPE=0 at the centre Accept for MP2 no net energy difference Condone missing 'electric' for MP2

Question	Marking guidance	Mark	Comments
04.5	No, it is not, because force/field strength/acceleration is not proportional to distance (from O) ✓	1	(instead it goes approximately like $-1/r^2$) Accept correct reference to Figure 8

Total		10	
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Question	Marking guidance	Mark	Comments
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14	B	10 V
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Suggestion is incorrect as Area between a and $X \gg$ area between X and b OR Idea that electric field is not uniform OR -1.5 kV would be potential at midpoint in a uniform field ✓ (Difference in) potential ΔV is area under graph (of E against r) ✓	No marks awarded if answer states suggestion is correct. Allow idea that more than half the area is between a and X Treat references to radial field as neutral Accept for MP2 for (idea that) $E = (-) \frac{\Delta V}{\Delta r}$	2	1 x AO1 1 x AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Evidence of use of $F = ma$ OR $2.7 \times 10^{-17} \times 91$ seen ✓ Evidence of their $F = EQ$ ✓ $E = 7.7 \times 10^3 \text{ (N C}^{-1}\text{)} \checkmark$	Expect to see $2.5 \times 10^{-15} \text{ (N)}$ Condone missing signs	3	1 x AO1 2 x AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	(Figure 6 shows) field strength decreases (away from rod) so (horizontal) force on particle decreases ✓ (horizontal component of) acceleration decreases ✓ Direction of force does not change so direction of acceleration does not change ✓	If no other mark is awarded, one mark can be given for idea that vertical component of acceleration is zero 2	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Use of work done = VQ OR $(-)\ 3000 \times (-)\ 3.2 \times 10^{-19}$ seen ✓ ₁ Initial KE = $\frac{1}{2}mv^2$ $= 3.6 \times 10^{-16} \text{ J}$ ✓ ₂ Calculation of their work done + their initial KE ✓ ₃ (Use of $\frac{1}{2}mv^2$ to give) $v = 9.9 \text{ m s}^{-1}$ ✓ ₄	Expect $9.6 \times 10^{-16} \text{ J}$ Expect to see $1.32 \times 10^{-15} \text{ J}$	4	AO3
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	D	from X to S	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Evidence of attempt to equate weight and force due to electric field ✓ $mg = EQ = Q\left(-\frac{V}{d}\right) \checkmark$	In MP1 it is sufficient to see mg replacing F in an equation where F is the force due to the electric field. In MP2 some indication of negative sign eg force due to electric field is in opposite direction to E or idea that $F_E + F_G = 0$ accept evidence from labels on diagram but 'oil drop is negative' is insufficient.	2	AO1 × 1 AO2 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Attempts to use $\frac{V}{d}$ to obtain field strength due to plates ✓ Attempts to use $E = \frac{Q_1}{4\pi\epsilon_0 r^2}$ for oil drop ✓ Finds difference between (magnitudes of) their MP2 their MP1 ✓ $8.7(5) \times 10^3 \text{ N C}^{-1}$ ✓ Direction consistent with difference in their two field strengths. ✓	Condone POT errors in MP1 and MP2. Only condone use of $d = 0.75 \text{ (mm)}$ for incorrect field strength due to plates. Expect to see $1.1 \times 10^4 \text{ N C}^{-1}$ Expect to see $1.9 \times 10^3 \text{ N C}^{-1}$ Accept $8.7 \times 10^3 \text{ N C}^{-1}$ Expect (E due to plates > E due to oil drop so direction is) downwards.	5	AO2 × 4 AO3 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Two from ✓✓ • upper plate less positive / lower plate more positive / decrease OR turn off pd across plates / reverse (polarity of) plates/ upper plate negative • increases separation of plates • (negative) charge on oil drop decreased / oil drop has lost (an) electron(s)/ oil drop is positive • idea that oil drop combines with an uncharged or positive drop or a drop with a lower specific charge		2	AO1
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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18	B	0	AO2
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22	A	r is the separation of the centres of the spheres.	AO1
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Total – 15 marks

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Use of $\frac{1}{2}mv^2$ OR Idea that $eV =$ their kinetic energy ✓ 620 or 623 or 624 (V) ✓	Accept 1.6×10^{-19} for e Accept more than 3sf answer that rounds to 623 or 624 Accept 622 using an e/m approach	2	1 × AO3 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Acceleration is constant / acceleration is (vertically) downwards ✓ Electric field is uniform/constant therefore the force on the electron is constant ✓ Idea that the force is downwards because the electron is negative OR that the force is downwards as it is in the opposite direction to the field. ✓	“Towards V_2 /bottom plate” is not enough for “downwards”.	3	1 × AO3 2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Evidence of 0.120 m divided by $1.48 \times 10^7 \text{ m s}^{-1}$ to give $8.1 \times 10^{-9} \text{ s}$ ✓	At least 2 sf	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO									
03.4	Three from: ✓✓✓ - determines the vertical displacement = 27 ± 1 (mm) - uses suvat with their displacement and time from 03.3 to find acceleration - uses $F = \text{their acceleration} \times \text{mass of electron}$ - uses their $F = \frac{V_2 e}{0.07}$ Answer 328 OR 330 (V) ✓	Alternative correct answers: <table><tr><td></td><td>8.1 ns</td><td>8.0 ns</td></tr><tr><td>26 mm</td><td>316 (V)</td><td>324 (V)</td></tr><tr><td>28 mm</td><td>340 (V)</td><td>349 (V)</td></tr></table> Allow third bullet point for combining $F = ma$ with $F = Eq$ or $F = \frac{Vq}{d}$ “Show that” value gives 336 or 340 V		8.1 ns	8.0 ns	26 mm	316 (V)	324 (V)	28 mm	340 (V)	349 (V)	4	2 × AO3 2 × AO2
	8.1 ns	8.0 ns											
26 mm	316 (V)	324 (V)											
28 mm	340 (V)	349 (V)											

Question	Answers	Additional comments/Guidelines	Mark	AO
03.5	Two from: ✓✓ - idea that the paths are broader as there is a range of electron speeds due to collisions with air particles - idea that the paths are fainter as some electrons do not reach the screen - idea that the paths bend more / y is greater as electrons are slower / so spend more time (accelerating)	Allow idea that the path is broader as electrons are scattered by collisions.	2	AO4

Total			12	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Use of $\Delta V = Ed$ ✓ 2.7×10^5 (V) ✓ 2 significant figures ✓	2.66×10^7	3	AO1

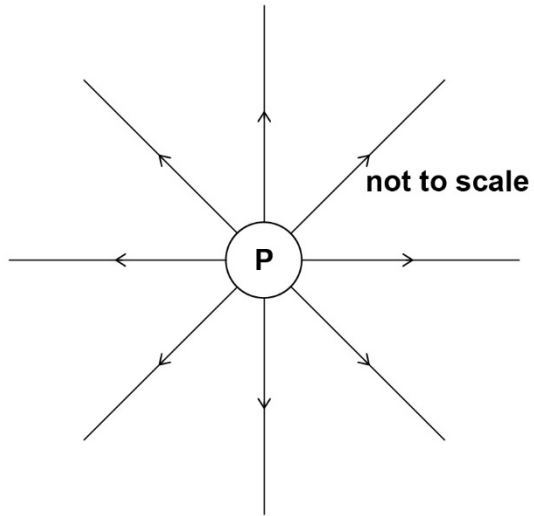
Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Use of $F = QE$ ✓ Use of $F = ma$ ✓ 1.27×10^{15} (m s ⁻²) ✓		3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	The electric field is 0 / $\Delta V = 0$ inside the cylinder ✓ So no work is done / there is no force ✓ Vacuum so no collisions with gas particles to slow electrons down / change their direction ✓		3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Protons travel faster in subsequent tubes ✓ Reference to $s = vt$ to give longer distance for same time ✓		2	AO2
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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11	D	9.0 nC	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Lines drawn which are straight and evenly spaced around the proton by eye with arrows away from the centre ✓	Allow lines not drawn with a ruler if they are straight and symmetrical by eye  not to scale Lines must be drawn connected to P for the mark to be awarded If there are additional lines drawn, for example equipotentials, then no marks given unless labelling for the field lines is clear.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Use of $F_E = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} \checkmark$ Use of $F_G = \frac{Gm_1 m_2}{r^2} \checkmark$ $2.3 \times 10^{39} \checkmark$	Condone the use of $\frac{kQq}{r^2}$ for mp1 but only if it rounds to 9×10^9 used for k $F_E = \frac{1}{4\pi\epsilon_0} \times \frac{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(5.29 \times 10^{-11})^2}$ $= 8.23 \times 10^{-8} \text{ N}$ $F_G = \frac{6.67 \times 10^{-11} \times 1.67 \times 10^{-27} \times 9.11 \times 10^{-31}}{(5.29 \times 10^{-11})^2}$ $= 3.63 \times 10^{-47} \text{ N}$ Ignore minus signs for forces or final answer $\frac{\text{electrostatic force}}{\text{gravitational force}} = \frac{8.23 \times 10^{-8}}{3.63 \times 10^{-47}}$ $= 2.3 \times 10^{39}$	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Any two from: ✓✓ - equating $\frac{mv^2}{r}$ and $\frac{1}{4\pi\epsilon_0} \frac{Q_1Q_2}{r^2}$ $v = 2.2 \times 10^6 \text{ (m s}^{-1}\text{)}$ - use of $E_k = \frac{1}{2}mv^2$ OR rearranging bp1 to equal $\frac{1}{2}mv^2$ $E_k = \frac{1}{2}mv^2 = \frac{Q_1Q_2}{8\pi\epsilon_0 r}$ $2.2 \times 10^{-18} \text{ (J)} \checkmark$	bp1 not awarded for alternative form of the electric force equation not on specification, eg $F = \frac{kQq}{r^2}$. $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Q_1Q_2}{r^2}$ $v = \sqrt{\frac{Q_1Q_2}{4\pi\epsilon_0 rm}}$ $v = \sqrt{\frac{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{4\pi\epsilon_0 \times 5.29 \times 10^{-11} \times 9.11 \times 10^{-31}}}$ $v = 2.19 \times 10^6 \text{ m s}^{-1}$ $E_k = \frac{1}{2} \times 9.11 \times 10^{-31} \times (2.19 \times 10^6)^2$ $E_k = 2.18 \times 10^{-18} \text{ J}$ Alternative approach Use of $\left(V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} \text{ and } W = QV \right)$ or $E_p = \frac{Q_1Q_2}{4\pi\epsilon_0 r}$ seen ✓ $E_k = (-) \frac{E_p}{2} \text{ used } \checkmark$ $2.2 \times 10^{-18} \text{ (J)} \checkmark$	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	Any two from: ✓✓ - uses $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ - uses $E_p = qV$ - uses $E_t = E_k + E_p$ and E_p is negative 2.2×10^{-18} (J) ✓	Condone the use of $V = \frac{kQ}{r}$ or $E_p = \frac{kQq}{r}$ for bp1 but only if it rounds to 9×10^9 used for k Use of $E_p = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r}$ gets bp1 and bp2 $V = \frac{1.6 \times 10^{-19}}{4\pi\epsilon_0 \times 5.29 \times 10^{-11}} = 27.2 \text{ V}$ $E_p = (-) 1.6 \times 10^{-19} \times 27.2 = (-) 4.35 \times 10^{-18} \text{ J}$ Allow ecf from 03.3 or use of 2.0×10^{-18} Allow 1 sf Alternative approach Allow energy required = E_k from 03.3 Condone 2×10^{-18} (J) (statements of energy required = E_k without a final answer is awarded 1 mark)	3	1 × AO1 2 × AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Unknown Unknown - Unit 3 - Q03

question	marking guidance	mark	comments
03.1	$t = \sqrt{\frac{2s}{g}}$ or $4.5 = \frac{1}{2} \times 9.81 \times t^2$ ✓ $t = 0.96 \text{ (s)}$ ✓	2	
03.2	Field strength = $186\,000 \text{ V m}^{-1}$ ✓ Acceleration = Eq / m or $186\,000 \times 1.2 \times 10^{-6}$ ✓ $0.22 \text{ (m s}^{-2}\text{)}$ ✓	3	
03.3	$0.10(3) \text{ (m)}$ (allow ecf from 03.2) ✓	1	

Unknown Unknown - Unit 3 - Q03 (continued 2)

03.4	Force on a particle = mg and acceleration = F / m so always = g ✓ Time to fall (given distance) depends (only) on the distance and acceleration ✓ OR: $g = GM / r^2$ ✓ Time to fall = $\sqrt{2s / g}$ so no m in equations to determine time to fall ✓	2	
03.5	Mass is not constant since particle mass will vary ✓ Charge on a particle is not constant ✓ Acceleration = Eq / m or $(V / d) (q / m)$ or Vq / dm ✓ E or V / d constant but charge and mass are 'random' variables so q / m will vary (or unlikely to be the same) ✓	3 max	

question	marking guidance	mark	comments
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10	D
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12	D
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